

US Minneapolis Metro Traffic Pattern Analysis

Objective:

- To Identify patterns, trends, and insights using Exploratory Data Analysis (EDA).
- To Clean and preprocess the dataset for analysis.
- To Create visualizations to explain traffic behavior.
- To Provide actionable insights for traffic management.

Outcome:

- The project identifies peak traffic periods, weather impacts, and traffic trends to support data-driven traffic management and planning.

Domain:

- Transportation / Traffic Analytics / Smart City Management

Dataset information:

Source: UCI Machine Learning

Repository [https://raw.githubusercontent.com/ManojKumarMaruthi/Regression/refs/head/master/Metro Interstate Traffic Volume.csv](https://raw.githubusercontent.com/ManojKumarMaruthi/Regression/refs/head/master/Metro%20Interstate%20Traffic%20Volume.csv)

Year / Timeline:

- Traffic data collected from 2012 to 2018
- Dataset published/updated in 2019 on UCI Repository

Dataset column/features description:

Column Name	Description
holiday	Public holiday indicator
temp	Temperature (Kelvin)
rain_1h	Amount of rainfall recorded in one hour
snow_1h	Amount of snowfall recorded in one hour
clouds_all	Cloud cover percentage
weather_main	Main weather condition
weather_description	Detailed weather condition
date_time	Date and time of observation
traffic_volume	Number of vehicles recorded

Tools used:

- Google Colab
- Python
- Power BI
- GitHub (Dataset Source)

Libraries used:

- **Pandas** – Used for data loading, cleaning, transformation, and analysis.
- **NumPy** – Used for numerical and mathematical operations.
- **Matplotlib** – Used for creating data visualizations and plots.
- **Seaborn** – Used for statistical data visualization and graphical analysis.
- **Plotly Express** – Used for creating interactive visualizations and charts.

Techniques applied:

- Data Collection
- Data Understanding
- Data Cleaning
- Missing Value Treatment
- Duplicate Record Analysis
- Feature Engineering
- Data Transformation
- Statistical Analysis
- Exploratory Data Analysis (EDA)
- Data Visualization
- Dashboard Development
- Business Insight Generation

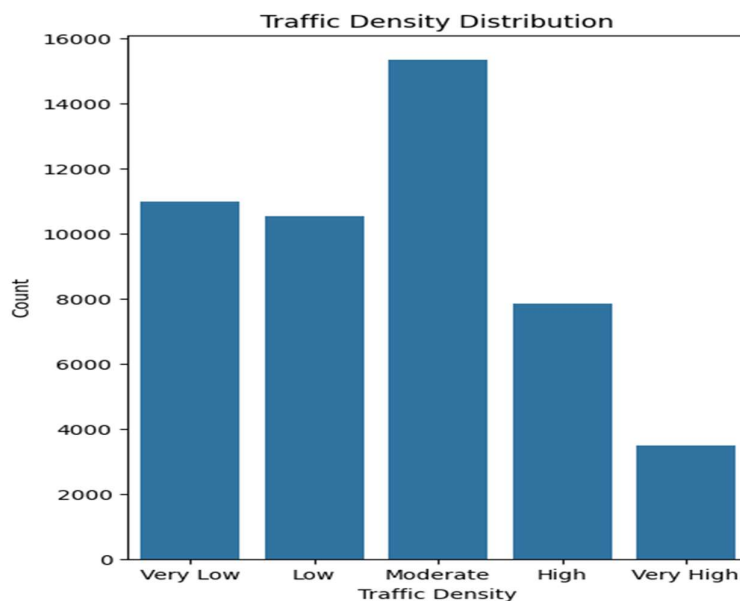
Features Implemented:

- Imported the Metro Interstate Traffic Volume dataset into Google Colab.
- Loaded the dataset into a Pandas DataFrame for analysis.
- Performed dataset exploration using shape, size, head, tail, info, and describe functions.
- Examined column names and data types to understand dataset structure.
- Identified and analyzed missing values in the dataset.
- Handled missing values in the Holiday column by replacing null values with "No Holiday".
- Checked for duplicate records and validated data quality.
- Renamed column names to improve readability and business understanding.
- Converted the Date_Time column into datetime format for time-based analysis.

- Performed statistical analysis using skewness and kurtosis to understand data distribution.
- Segregated numerical and categorical features for further analysis.
- Created a new feature called Traffic Density based on Traffic Volume ranges.
- Categorized traffic levels into Very Low, Low, Moderate, High, and Very High.
- Validated the cleaned dataset after preprocessing.
- Saved the cleaned dataset as a CSV file.
- Downloaded the processed dataset for Power BI integration.
- Conducted Exploratory Data Analysis (EDA) to identify traffic patterns and trends.
- Performed weather-based traffic analysis using rainfall, snowfall, temperature, and cloud cover data.
- Analyzed traffic distribution across different traffic density categories.
- Generated visualizations to understand relationships between traffic volume and influencing factors.
- Prepared the final dataset for dashboard creation and business insights generation.
- Integrated the cleaned dataset into Power BI for reporting and visualization.

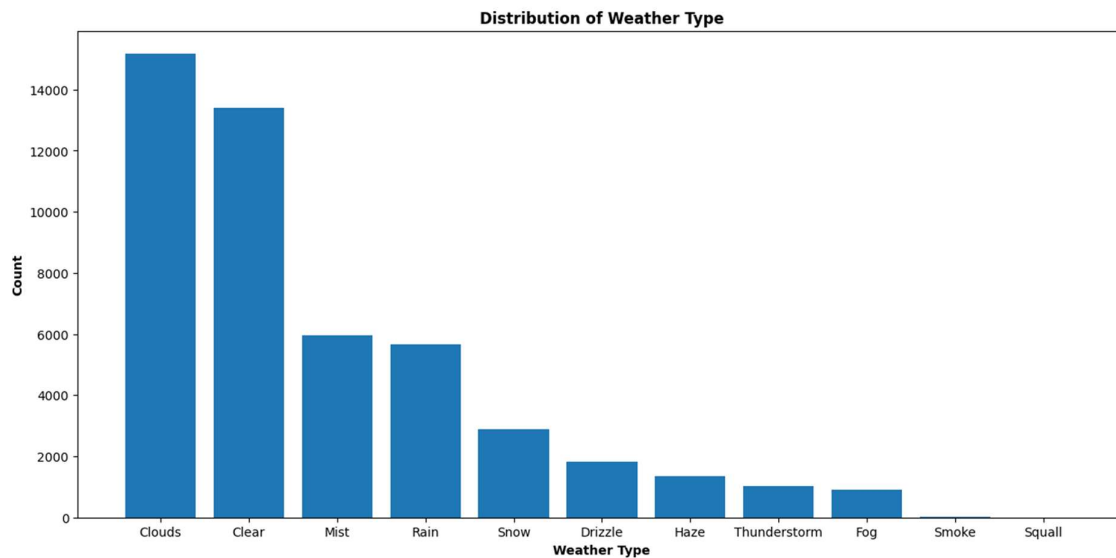
EDA and Visualizations:

Bar Chart:



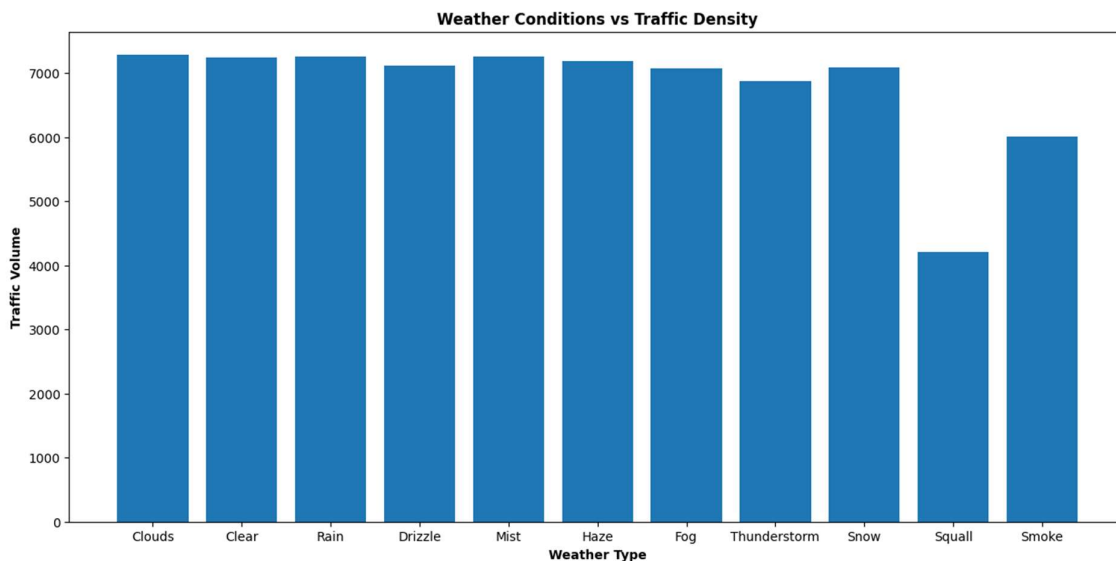
Insights:

- Moderate traffic density is the most common category, with approximately 15,300 records.
- Very Low and Low traffic densities are also frequent, with around 11,000 and 10,500 records respectively.
- High traffic density occurs less often, accounting for about 7,800 records.
- Very High traffic density is the least frequent category, with only about 3,500 records.



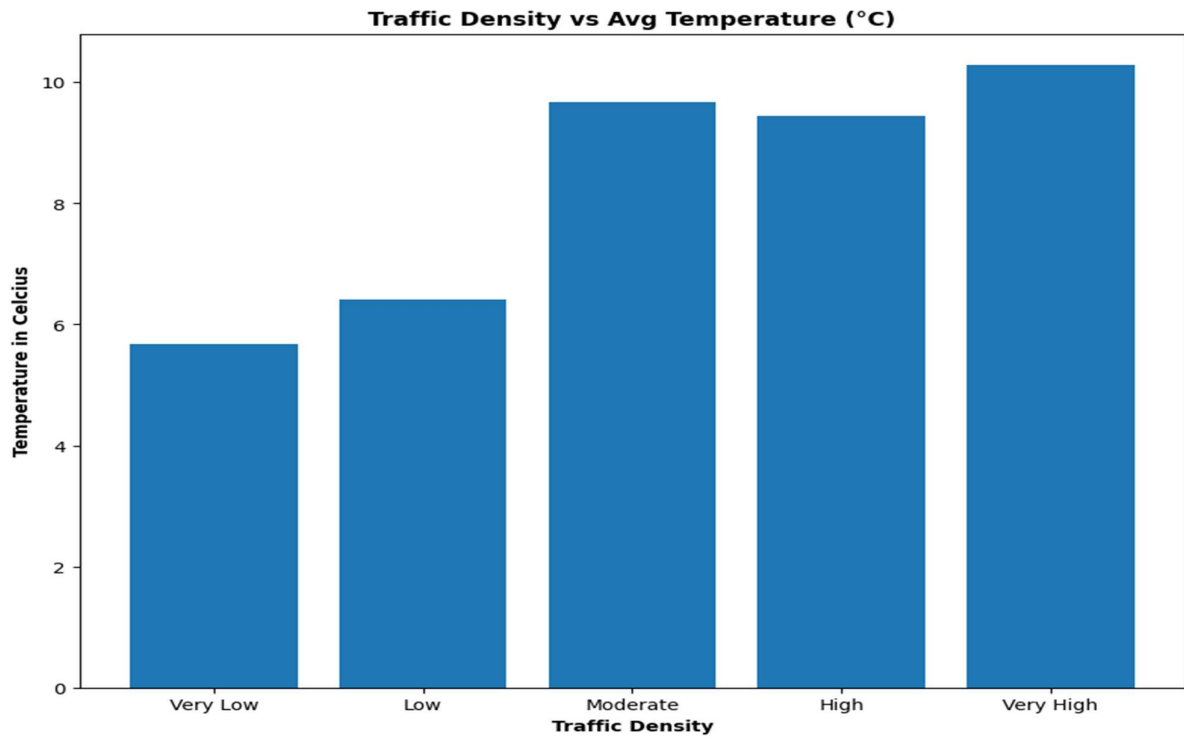
Insights:

- Cloudy weather is the most common weather condition in the dataset.
- Clear weather is also very common, showing that most days had normal weather conditions.
- Mist and Rain occur sometimes, but much less often than Cloudy and Clear weather.
- Snow, Fog, Thunderstorms, Smoke, and Squall are rare, meaning extreme weather conditions happened only a few times



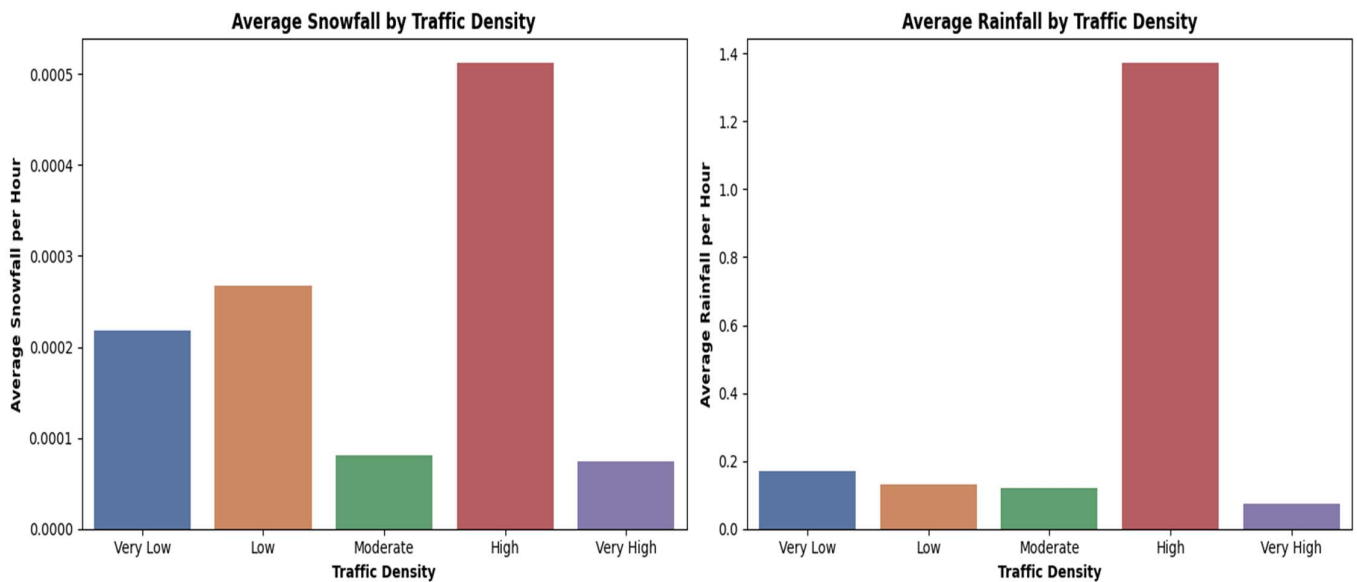
Insights:

- Traffic volume remains high across most weather conditions, showing that weather has only a limited impact on overall traffic flow.
- Cloudy, Clear, Rainy, and Misty weather have similar traffic volumes, indicating that people continue to travel even during light rain or reduced visibility.
- Haze records the highest traffic volume, while Thunderstorms show a slightly lower traffic volume compared to other common weather conditions.
- Squall and Smoke have the lowest traffic volumes, suggesting that severe or unusual weather conditions may reduce traffic movement.



Insights:

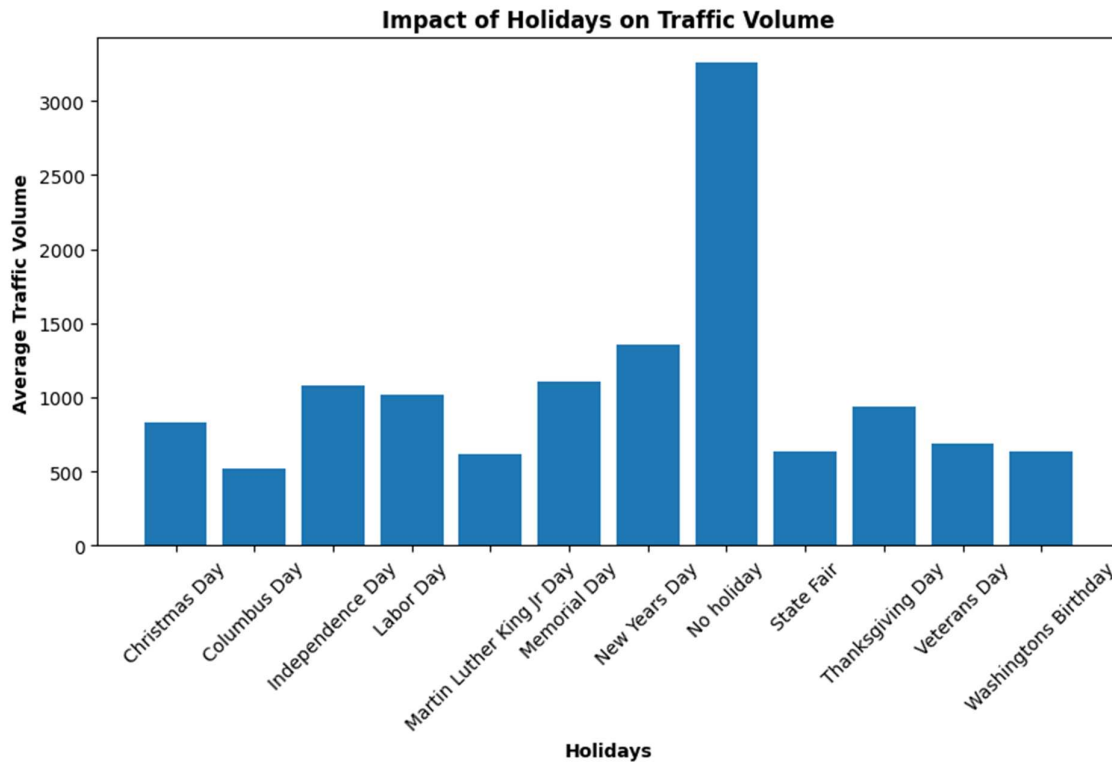
- Average temperature increases as traffic density increases.
- Very High traffic density has the highest average temperature (around 10°C).
- Very Low traffic density has the lowest average temperature (around 6°C).
- Moderate, High, and Very High traffic levels are associated with relatively warmer temperatures.



Insights:

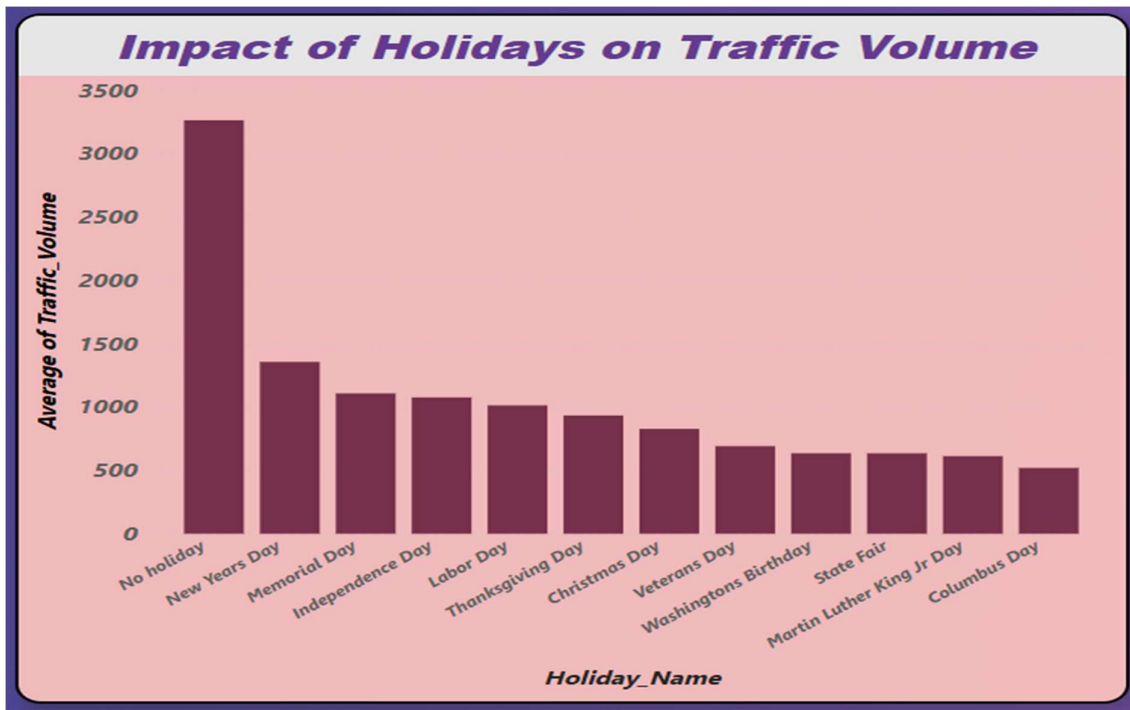
- High traffic density has the highest average rainfall, significantly higher than all other categories.
- Very Low, Low, and Moderate traffic densities have similar rainfall levels.
- Very High traffic density shows the lowest average rainfall.

- Rainfall appears to have a noticeable impact on traffic, especially during High traffic conditions.
- High traffic density records the highest average snowfall per hour.
- Moderate and Very High traffic densities have the lowest snowfall levels.
- Low and Very Low traffic densities show moderate snowfall.
- No clear linear relationship exists between traffic density and snowfall.
- High traffic is a noticeable outlier with significantly higher snowfall.



Insights:

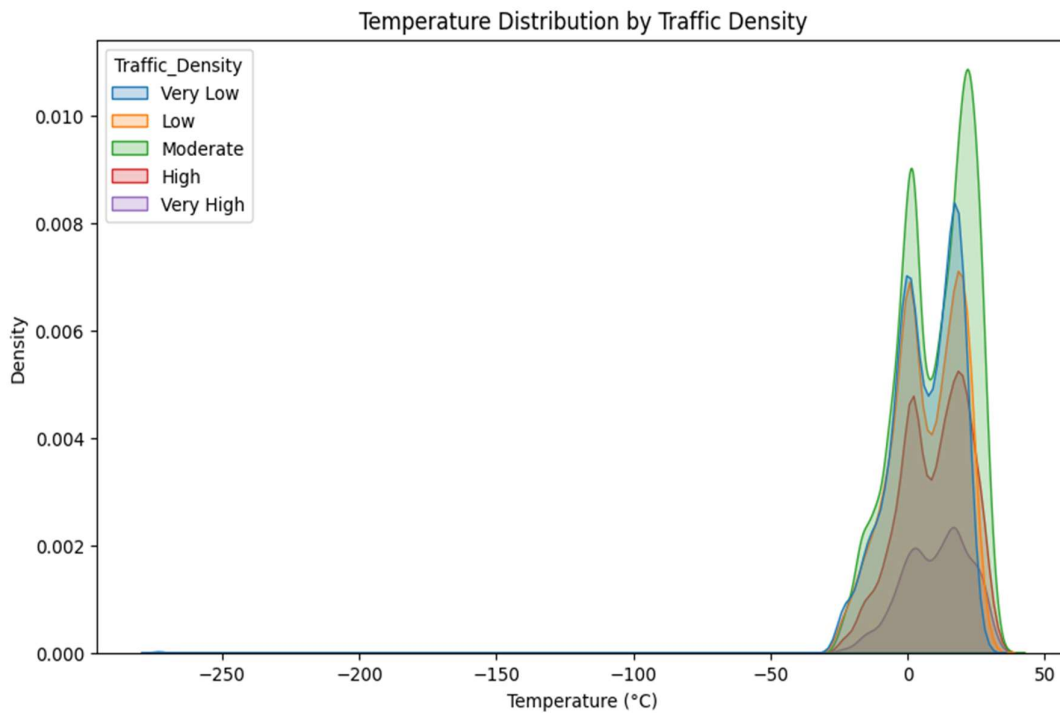
- Traffic volume is highest on non-holidays, significantly exceeding all holiday periods.
- New Year's Day records the highest traffic among holidays.
- Memorial Day, Independence Day, and Labor Day show relatively high traffic volumes.
- Columbus Day has the lowest average traffic volume.
- Most holidays experience reduced traffic compared to non-holiday days, indicating a holiday effect on traffic patterns.



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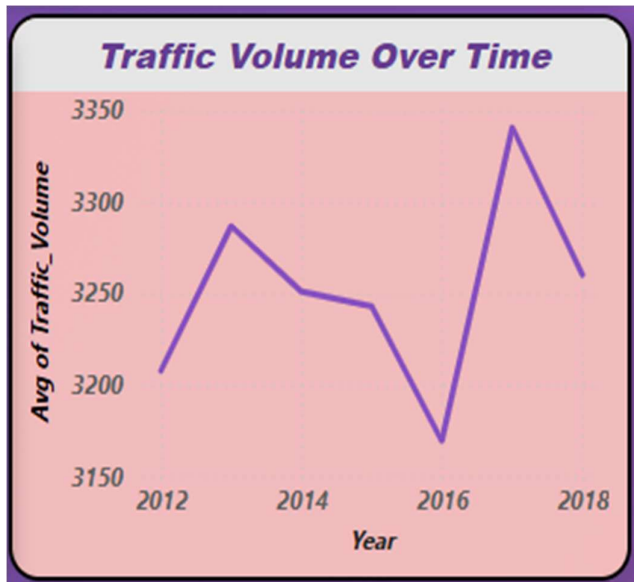
Area chart:



Insights:

- Most temperature values are between 0°C and 25°C across all traffic density levels.
- Moderate traffic density shows the highest concentration of observations.
- Temperature distributions are similar for all traffic density categories.
- A few extreme outliers are present and should be investigated before removal.

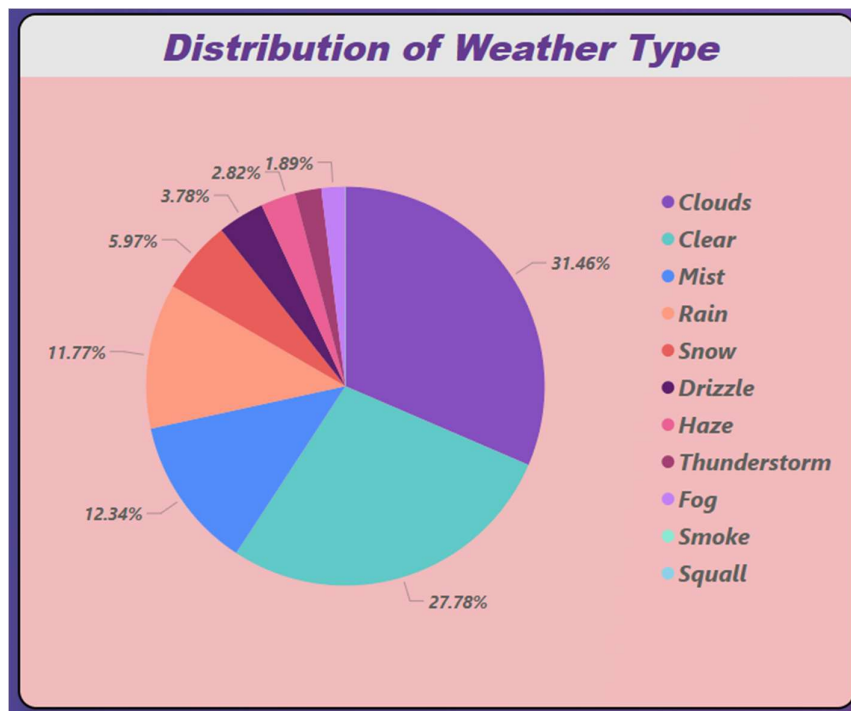
Line Chart:



Insights:

- Traffic volume remains consistently high throughout the years, with most values ranging between 5,000 and 7,000 vehicles.
- Traffic levels show regular fluctuations over time, indicating variations in daily and hourly traffic patterns.
- There is no significant upward or downward trend in traffic volume, suggesting that overall traffic demand remained stable during the observed period.
- A gap in the timeline around 2014–2015 suggests the presence of missing data or a period when traffic observations were not recorded.

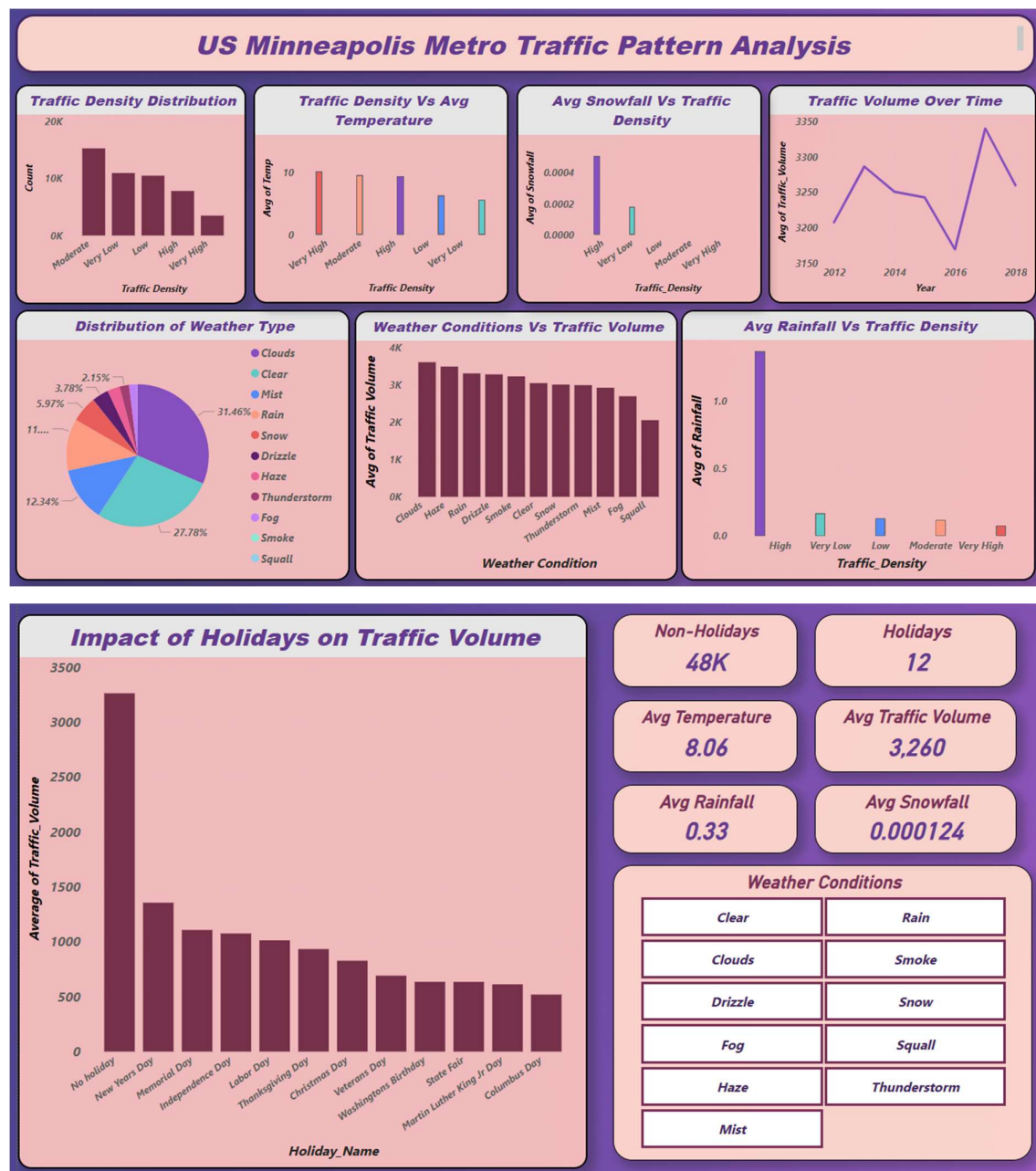
Pie Chart:



Insights:

- Cloudy weather is the most common condition, accounting for 31% of observations.
- Clear weather is the second most frequent, contributing 28%.
- Mist and Rain each make up 12%, indicating moderate occurrence.
- Snow accounts for 6%, while Drizzle (4%) and Haze (3%) are less common.
- Thunderstorms and Fog are relatively rare, each representing 2% of the data.
- Smoke has a negligible occurrence (0%), making it the least frequent weather condition.
- Nearly 60% of the observations occur under clear or cloudy conditions, suggesting that normal weather dominates the dataset.

Dashboards:



Future Enhancement:

- Integrate real-time traffic data for live monitoring and analysis.
- Incorporate GPS and route-level traffic information.
- Develop machine learning models for traffic volume forecasting.
- Build automated congestion prediction and alert systems.
- Include accident, roadwork, and event-based traffic impact analysis.
- Create interactive Power BI dashboards with advanced drill-through capabilities.
- Implement geospatial visualizations for location-specific traffic analysis.
- Integrate weather forecast data for proactive traffic management.

Conclusion:

- Successfully cleaned, transformed, and analyzed the Metro Traffic dataset.
- Identified key traffic patterns and congestion trends across different traffic density levels.
- Examined the impact of weather conditions such as rainfall and snowfall on traffic volume.
- Created meaningful visualizations and an interactive Power BI dashboard for better decision-making.
- The analysis provides valuable insights that can support traffic management and transportation planning.
- The project demonstrates how data analytics can be used to understand and improve traffic operations.

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